Land Use Edge Weights via NLCD: Residential-Commercial Transition Detection

Giovanni Della-Libera

June 28, 2026

Abstract

Physical land use—the actual pattern of residential neighborhoods, commercial corridors, industrial zones, open land, and water bodies on the ground—is among the oldest and most intuitive bases for community of interest recognition in American redistricting. Yet algorithmic redistricting systems have historically operationalized geographic community only through boundary length, ignoring the land cover character of the tracts on either side of each boundary. This paper presents M.2, the second paper in the M-track community character series, which operationalizes physical land use through the USGS National Land Cover Database (NLCD) 2021 30-meter raster.

For each census tract we compute a four-fraction land use character vector: fraction of pixels classified as residential (NLCD classes 22–23), commercial and industrial (class 24), open developed (class 21), and natural (all other non-water classes including forest, wetland, shrub, and agricultural). Edge weights between adjacent tracts are modulated by the cosine similarity of their land use vectors: residentially similar tracts receive heavier edges; residential-to-commercial transitions receive lighter edges that serve as natural district cut seams. A hard-cut rule enforces water boundaries: any adjacency where both tracts have majority water classification (class 11) receives an edge weight of zero, preventing the algorithm from assigning cross-water tracts to the same district in the absence of explicit bridge adjacency.

Experimental design targets North Carolina ($k = 14$), Wisconsin ($k = 8$), and Texas ($k = 38$) with the primary metric of fraction of district boundaries coinciding with NLCD land-use class transitions. Compactness comparison (Polsby-Popper) is the secondary metric. All quantitative results are deferred to Phase 2 pending NLCD GeoTIFF download and zonal statistics computation. The NLCD data is annual, free, publicly available, and contains no racial or partisan content, making physical land use one of the most legally defensible community-of-interest signals available.

Keywords

Land use, NLCD, Physical geography, Communities of interest, Edge weights, METIS, Water boundary, Hard cut, Redistricting

1 Introduction

When the Ohio Redistricting Commission drew state legislative maps in 2021, residents of the Cuyahoga River valley testified that their community identity was shaped more by the physical landscape—the valley floor, the bluffs, the river itself—than by county or municipal boundaries. The Commission noted the testimony and then, lacking any computational mechanism to encode physical land use as a community signal, drew lines based on population equality and VRA compliance. The valley communities were divided across three districts that reflected none of the physical geography that defined them.

This paper provides that computational mechanism.

Physical land use differs from administrative community signals (M.6 zone co-membership) and economic community signals (M.1 economic character) in an important respect: it is what actually exists on the ground. Zoning maps reflect what a municipality *permits*; land use maps reflect what has actually been built. A census tract may be zoned for mixed use but be entirely developed as a residential neighborhood; the NLCD classification captures the residential reality, not the zoning aspiration. For redistricting purposes—where the question is what kind of community actually lives within a proposed district—the physical reality is the appropriate input.

The M-track community character framework, introduced in M.0 [Della-Libera, 2025], operationalizes diverse community signals as edge weights in the census-tract adjacency graph. Communities that belong together receive heavy edges that METIS avoids cutting; communities that are naturally distinct receive light edges that invite cuts at community transitions. M.2 is the land use implementation paper. It applies the M-track framework using the USGS National Land Cover Database (NLCD) 2021, the most comprehensive and authoritative land use classification of the contiguous United States.

Physical land use is the right M-track signal for several reasons.

1. **Empirical grounding.** NLCD is a remote sensing product derived from Landsat satellite imagery. It measures what is actually on the ground, not what jurisdictions say should be there. Classification accuracy for the developed land classes is above 85% based on the MRLC validation protocol [Homer et al., 2020, Dewitz and U.S. Geological Survey, 2023].

2. **Legal recognition.** Physical geography—including natural boundaries such as rivers, mountain ridges, and coastal features—is one of the oldest recognized redistricting criteria. Courts have consistently upheld plans that follow natural boundaries and struck down plans that cross natural boundaries without geographic justification.

3. **Water boundary doctrine.** The hard-cut rule for majority-water adjacencies (discussed in Section 3) implements a well-established legal doctrine: that water bodies are natural community boundaries and that cross-water districts require specific justification.

4. **Partisan neutrality.** No electoral, racial, or demographic data enters the computation. The signal is the physical character of the land surface.

Distinction from M.3 housing character. The M.3 paper [Della-Libera, 2026a] operationalizes housing character through ACS survey data on housing tenure, age, and value. M.2 operationalizes the physical structure of what is built. The signals overlap—residential development concentration (M.2) correlates with single-family housing density (M.3)—but are not redundant. M.2 captures the spatial pattern of development types as seen from satellite; M.3 captures the socioeconomic character of the households living in that development. Both contribute to the M.8 composite index [Della-Libera, 2026b].

Paper organization. Section 2 describes the NLCD 2021 data and the physical geography of communities of interest. Section 3 defines the land use character vector, cosine similarity, hard-cut rule, and edge weight formula. Section 4 describes the experimental design and defers quantitative results to Phase 2. Section 5 provides a legal usage guide including the water boundary doctrine and natural boundary case law. Section 6 concludes.

2 Background and Data

2.1 NLCD 2021: Data and Coverage

The National Land Cover Database (NLCD) is produced by the Multi-Resolution Land Characteristics (MRLC) Consortium, a partnership of federal agencies including the USGS, EPA, USDA Forest Service, and National Park Service [Homer et al., 2020]. NLCD provides a consistent, wall-to-wall classification of land cover for the contiguous United States at 30-meter spatial resolution, derived from Landsat 8 satellite imagery supplemented by ancillary datasets (building footprints, impervious surface maps, canopy cover estimates).

The 2021 vintage is the most recent NLCD release as of 2024. It is available as a single GeoTIFF covering CONUS (approximately 2 GB compressed), in Albers Equal Area projection (EPSG:5070), with 30-meter pixel resolution. The data is published at `https://www.mrlc.gov/data/nlcd-2021-land-cover-conus` and is public domain under U.S. government work provisions.

NLCD class hierarchy. NLCD uses a hierarchical 16-category system derived from the Anderson Level II classification [Anderson et al., 1976]. Table 1 lists all classes with their integer codes and the M.2 category assignment.

Table 1: NLCD 2021 land cover classes and M.2 category assignments. Class 11 is excluded from the cosine similarity vector and used only for the hard-cut rule.

Code	NLCD class name	M.2 category
11	Open Water	Hard-cut trigger (not in vector)
21	Developed, Open Space	`pct_open`
22	Developed, Low Intensity	`pct_residential`
23	Developed, Medium Intensity	`pct_residential`
24	Developed, High Intensity	`pct_commercial`
31	Barren Land	`pct_natural`
41	Deciduous Forest	`pct_natural`
42	Evergreen Forest	`pct_natural`
43	Mixed Forest	`pct_natural`
52	Shrub/Scrub	`pct_natural`
71	Grassland/Herbaceous	`pct_natural`
81	Pasture/Hay	`pct_natural`
82	Cultivated Crops	`pct_natural`
90	Woody Wetlands	`pct_natural`
95	Emergent Herbaceous Wetlands	`pct_natural`

The NLCD "developed" classes (21–24) form a gradient of impervious surface density. Class 21 (developed, open space, $<$20% impervious) includes suburban lawns, golf courses, parks in developed areas, and cemeteries. Class 22 (low intensity, 20–49% impervious) is the predominant classification for single-family residential neighborhoods. Class 23 (medium intensity, 50–79% impervious) captures denser residential development: row houses, townhouses, multi-family housing, and mixed-use residential buildings. Class 24 (high intensity, $\geq$80% impervious) captures commercial districts, industrial parks, transportation hubs, and urban core blocks where nearly the entire land surface is covered by buildings, pavement, or other impervious materials.

Zonal statistics computation. NLCD is a raster product; census tracts are polygon features from TIGER/Line shapefiles [U.S. Census Bureau, 2020]. Computing land use character vectors requires zonal statistics: for each tract polygon, count the number of 30-meter pixels of each NLCD class within the polygon boundary.

The computation proceeds as follows:

1. Reproject TIGER/Line tract polygons from EPSG:4326 to EPSG:5070 (NLCD native projection) to avoid reprojection artifacts in the raster sampling.
2. For each tract, use the GDAL `zonal_stats` function to count pixels within the polygon boundary (all-touched: false; pixel center must fall within polygon).
3. Aggregate pixel counts by M.2 category to produce five per-tract counts: `n_residential`, `n_commercial`, `n_open`, `n_natural`, `n_water`.
4. Divide by total pixel count to produce fractions.

Edge cases. Census tracts in coastal areas or along large rivers may have substantial water pixels (class 11) within their polygons even if the residential area of the tract is inland. The water fraction is excluded from the cosine similarity vector to prevent high water pixel counts from suppressing similarity between two inland residential tracts that happen to border the same body of water. Instead, water is handled by the hard-cut rule described in Section 3.

2.2 Physical Geography as Communities of Interest

The use of physical geography as a redistricting criterion predates the modern communities-of-interest doctrine. The first American redistricting practitioners drew district lines along rivers, mountain ridges, and coastal features because those were the natural divisions visible on maps and meaningful to the communities they were drawing.

Schwartzberg [1966] noted the relationship between geographic compactness and natural land use boundaries: compact districts tend to follow natural divisions, and natural divisions tend to reflect the physical landscape that shapes community identity. The connection is not merely aesthetic. Communities that share a physical landscape share infrastructure needs (flood management, shoreline erosion, wildfire risk, agricultural water management) that translate directly into shared legislative interests.

Urban-rural transitions. The transition from residential suburban development to commercial corridor, or from developed land to agricultural land, is one of the most consistent physical geographic community boundaries in American metropolitan areas. Residents of a suburban neighborhood and workers in an adjacent industrial park may share a common boundary on a map but live in economically and physically distinct communities. The NLCD classification makes this distinction explicit: one side of the boundary is class 22 (low-intensity residential), the other is class 24 (high-intensity developed, commercial/industrial).

Water bodies as community barriers. Rivers, bays, sounds, and lakes create natural community boundaries by imposing travel barriers and defining watershed communities with shared environmental concerns. North Carolina's Pamlico Sound divides the coastal mainland from the Outer Banks, creating communities whose access, infrastructure, and economic character are fundamentally different. Louisiana's bayou network defines communities more precisely than parish boundaries. Texas's Gulf Coast bays and barrier islands define distinct communities not well captured by county lines. The hard-cut rule in M.2 encodes this physical reality as a zero-weight edge: communities separated by majority-water adjacencies are prevented from being placed in the same district unless explicit bridge adjacency is recorded in the graph structure.

3 Algorithm

3.1 Land Use Character Vector

The M-track framework (M.0) defines community character as a function $\mathbf{c} : \mathcal{T} \to \mathbb{R}^k_{\geq 0}$ mapping each census tract to a non-negative character vector [Della-Libera, 2025]. For M.2, the character function is the land use fraction vector:

$$\mathbf{l}(t) = \Big(p_{\text{res}}(t),\ p_{\text{com}}(t),\ p_{\text{open}}(t),\ p_{\text{nat}}(t)\Big) \tag{1}$$

where the components are the fractions of NLCD pixels within tract t belonging to each M.2 category.

Component definitions.

$$p_{\text{res}}(t) = \frac{n_{22}(t) + n_{23}(t)}{N(t)} \tag{2}$$

$$p_{\text{com}}(t) = \frac{n_{24}(t)}{N(t)} \tag{3}$$

$$p_{\text{open}}(t) = \frac{n_{21}(t)}{N(t)} \tag{4}$$

$$p_{\text{nat}}(t) = \frac{n_{31}(t) + n_{41}(t) + \cdots + n_{95}(t)}{N(t)} \tag{5}$$

where $n_k(t)$ is the count of class-k pixels within tract t, and $N(t) = n_{21}(t) + n_{22}(t) + n_{23}(t) + n_{24}(t) + n_{31}(t) + \cdots + n_{95}(t)$ is the total non-water pixel count. Water pixels (class 11) are excluded from the denominator $N(t)$.

The four-component vector sums to 1.0 by construction: $p_{\text{res}} + p_{\text{com}} + p_{\text{open}} + p_{\text{nat}} = 1$. This normalization ensures that the cosine similarity between two purely residential tracts ($\mathbf{l} = (1, 0, 0, 0)$) is 1.0 regardless of their pixel density, which is the correct behavior: two densely developed residential neighborhoods are more similar to each other than either is to a commercial zone, regardless of how many 30-meter pixels each contains.

Water fraction. The per-tract water fraction is computed separately:

$$p_{\text{wat}}(t) = \frac{n_{11}(t)}{n_{11}(t) + N(t)} \tag{6}$$

and is stored alongside the four-component vector but excluded from the cosine similarity computation. It is used only for the hard-cut rule.

3.2 Cosine Similarity

The community similarity between adjacent tracts u and v is the cosine similarity of their land use character vectors:

$$\text{sim}(\mathbf{l}_u, \mathbf{l}_v) = \frac{\mathbf{l}_u \cdot \mathbf{l}_v}{||\mathbf{l}_u|| \, ||\mathbf{l}_v||} \tag{7}$$

Table 2 shows the character vectors for five prototypical tract types encountered in American metropolitan areas.

Note that commercial corridor and industrial park have nearly identical vectors under the M.2 classification, because NLCD class 24 (high-intensity developed) does not distinguish commercial from industrial use at 30-meter resolution. The distinction between these economic character types is better captured by M.1 (LODES economic character) than by M.2.

Table 2: Prototypical land use character vectors for five tract types. All fractions sum to 1.0.

Type	p_{res}	p_{com}	p_{open}	p_{nat}
Dense residential suburb	0.75	0.03	0.18	0.04
Strip-mall commercial corridor	0.05	0.85	0.08	0.02
Industrial park / logistics	0.02	0.88	0.08	0.02
Exurban open space	0.12	0.01	0.35	0.52
Rural agricultural	0.01	0.00	0.10	0.89

M.2's comparative strength is the residential-to-non-residential transition and the natural land cover spectrum, not the internal differentiation of developed zones.

Table 3 shows the pairwise cosine similarity between the five prototypical vectors.

Table 3: Cosine similarity between prototypical land use tract types. Values near 1.0 indicate strong community similarity; values near 0.0 indicate natural cut seams.

	Dense res.	Commercial	Industrial	Exurban	Rural
Dense residential	1.00	0.12	0.09	0.38	0.08
Commercial corridor	0.12	1.00	0.99	0.11	0.04
Industrial park	0.09	0.99	1.00	0.10	0.03
Exurban open space	0.38	0.11	0.10	1.00	0.88
Rural agricultural	0.08	0.04	0.03	0.88	1.00

The similarity matrix reveals the geographic community structure:

- **Residential to commercial (0.12).** The sharpest community boundary in the M.2 framework is the edge between a residential neighborhood and an adjacent commercial corridor. This is the natural cut seam M.2 is designed to exploit.

- **Commercial to industrial (≈0.99).** NLCD class 24 dominates both; the two tract types are nearly indistinguishable from satellite imagery. M.1 (economic character) is the appropriate tool for distinguishing these.

- **Rural to exurban (0.88).** A soft similarity; both are primarily natural land cover. The gradient from agricultural to exurban residential is gentler than the urban residential-to-commercial transition.

- **Residential to rural (0.08).** A moderate hard boundary between the developed suburban fabric and agricultural or forested land: another strong M.2 cut seam.

3.3 The Hard-Cut Rule for Water Boundaries

The M.2 algorithm imposes a hard override on any adjacency where both adjacent tracts have majority water classification:

$$\text{if } \min\Big(p_{\text{wat}}(u),\, p_{\text{wat}}(v)\Big) > 0.5 \quad \Rightarrow \quad w(u,v) = 0 \tag{8}$$

This rule is applied *before* the cosine similarity computation and takes absolute priority. Even if two water-adjacent tracts have identical land use vectors, the edge weight between them is forced to zero, preventing METIS from assigning them to the same district unless an explicit bridge adjacency overrides the rule.

The threshold of 0.5 is conservative by design. A tract must be majority water for the hard-cut rule to activate. Tracts with 30–49% water pixels (common in coastal areas and along rivers) do not trigger the hard cut; instead, their water fraction suppresses the non-water pixel counts, resulting in a naturally lower cosine similarity with inland tracts.

Bridge adjacency override. The BISECT adjacency graph can record explicit bridge adjacency edges in addition to contiguity-derived edges. For coastal geographies where a bridge or causeway physically connects two majority-water tracts, the graph constructor can add a bridge edge with a user-specified weight. The hard-cut rule applies only to edges derived from polygon contiguity; bridge edges are not subject to the water cut override.

3.4 Edge Weight Formula

The M.2 edge weight follows the M-track blend formula:

$$w_{\text{lu}}(u,v) = \begin{cases} 0 & \text{if hard-cut rule (8) applies} \\ w_{\text{geo}}(u,v) \times \Big(\alpha + (1-\alpha)\,\text{sim}(\mathbf{l}_u, \mathbf{l}_v)\Big) & \text{otherwise} \end{cases} \tag{9}$$

with default $\alpha = 0.5$. The blend factor α prevents the weight from collapsing to zero for dissimilar non-water adjacent tracts, preserving geographic boundary length as a baseline constraint.

CLI invocation.

```
bisect state --state NC --year 2020 \
    --weights-override land-use \
    --version v_landuse_test
```

The `land-use` weight mode reads precomputed NLCD fraction vectors from the bisect data cache. Vectors are computed offline using the zonal statistics pipeline described in Section 2 and stored in Parquet format alongside the adjacency graph.

3.5 Test Invariants

The following L0 unit test invariants must hold for any correct implementation:

`water_majority_gives_zero_weight`. For any tract pair (u, v) where $p_{\text{wat}}(u) > 0.5$ and $p_{\text{wat}}(v) > 0.5$: $w_{\text{lu}}(u, v) = 0.0$.

`identical_landuse_gives_max_weight`. For any two tracts with identical land use fraction vectors $\mathbf{l}_u = \mathbf{l}_v$: $\text{sim}(\mathbf{l}_u, \mathbf{l}_v) = 1.0$ and $w_{\text{lu}}(u, v) = w_{\text{geo}}(u, v)$ (blend reduces to $\alpha + (1 - \alpha) = 1$).

`residential_vs_commercial_weight_low`. For a tract pair where $p_{\text{com}}(u) \geq 0.9$ and $p_{\text{res}}(v) \geq 0.9$: $w_{\text{lu}}(u, v) < 0.3 \times w_{\text{geo}}(u, v)$ (using $\alpha = 0.0$ for this test).

4 Empirical Design

4.1 Experimental Design

Quantitative results for M.2 land use weights are deferred to Phase 2, pending download and zonal statistics computation from the NLCD 2021 GeoTIFF.[1] This section describes the experimental protocol that will govern Phase 2 execution.

States. Three states are selected to represent distinct land use geography profiles:

- **North Carolina ($k = 14$).** NC provides coastal water complexity (Outer Banks, Pamlico Sound, Albemarle Sound), inland development transition zones (Research Triangle residential-commercial edge), and agricultural land in the Coastal Plain. The hard-cut rule is expected to produce tangible effects on the Pamlico Sound adjacencies where mainland tracts and Outer Banks tracts are technically contiguous through narrow water passages.

[1] Phase 2 will be triggered after the NLCD 2021 GeoTIFF is downloaded to the bisect data cache. The data URL is `https://www.mrlc.gov/data/nlcd-2021-land-cover-conus`. Estimated download size: ~2 GB compressed.

- **Wisconsin ($k = 8$).** WI provides a test for diffuse land use patterns. Milwaukee's urban core has a defined commercial-residential transition; the rest of the state is a mixture of agricultural land, small cities, and forest. This provides a partial contrast to NC: the hard-cut rule may affect Green Bay coastal adjacencies; the cosine similarity signal may have modest effect in Milwaukee but not statewide.
- **Texas ($k = 38$).** TX provides the full land use diversity spectrum: Gulf Coast water boundaries (Galveston Bay, Matagorda Bay, Corpus Christi Bay), dense urban commercial corridors (Houston Loop 610, Dallas Galleria), extensive agricultural and ranchland, and desert/scrub in West Texas. The hard-cut rule is expected to affect numerous coastal and bay adjacencies. The cosine similarity signal may produce strong effects in the Houston and Dallas commercial transition zones.

All experiments use:

- Algorithm: `standard-bisect` structure
- Seed: 42 (single seed; multi-seed variance designated Phase 3)
- Blend factor: $\alpha = 0.5$ (`-land-alpha 0.5`)
- Year: 2020 decennial census population
- Comparison baseline: `-weights-override geographic` on identical plan configuration

4.2 Primary Metric: Land-Use Boundary Coincidence

The M-track primary metric for M.2 is the *fraction of district boundaries coinciding with NLCD land-use class transitions.*

A district boundary segment is said to coincide with a land-use transition if the two tracts separated by that boundary segment have cosine similarity $\text{sim}(\mathbf{l}_u, \mathbf{l}_v) < 0.4$ under the M.2 vector definition. Equivalently: the boundary is at a point where the physical land use type changes substantially (residential-to-commercial, developed-to-natural, or natural-to-agricultural).

Formally:

$$\text{LUB} = \frac{\#\{(u, v) \in \partial(\pi) : \text{sim}(\mathbf{l}_u, \mathbf{l}_v) < 0.4\}}{\#\partial(\pi)} \tag{10}$$

where $\partial(\pi)$ denotes the set of cut edges in the plan π. A higher LUB score means district boundaries fall more consistently at natural physical transitions, indicating higher physical community coherence.

The prediction for Phase 2:

- LUB under land-use weights $\geq$ LUB under geographic weights (land-use weights are specifically designed to prefer cuts at transitions).
- The improvement in LUB should be at least 20 percentage points on NC-14, where the coastal and development-transition geographies are well-suited to the signal. This target is marked †Phase 2.

4.3 Secondary Metric: Compactness

Polsby-Popper compactness [Polsby and Popper, 1991] is the secondary metric. The prediction is that land-use weights should not reduce compactness: cutting at physical transition zones tends to produce rounder district shapes because transition zones are often linear features (shorelines, commercial strips, ridge lines) that serve as cleaner boundary segments than arbitrary internal boundaries.

Formally, the M.2 compactness prediction:

$$\overline{PP}_{\text{landuse}} \geq \overline{PP}_{\text{geographic}} \tag{11}$$

where $\overline{PP}$ is mean Polsby-Popper across all districts in the plan. This prediction will be confirmed or refuted in Phase 2.†

4.4 Hard-Cut Validation

An additional validation metric is specific to the hard-cut rule:

> *Zero districts in the final plan cross a majority-water edge as defined by equation* (8).

This is a hard constraint, not a soft metric. Any plan produced by `bisect` with `-weights-override land-use` must satisfy this constraint by construction (the hard-cut weight of zero means METIS cannot assign both endpoints of a water edge to the same partition component under the minimum-cut objective). Phase 2 will verify this property on the NC Pamlico Sound adjacencies and the Texas Gulf Coast bay adjacencies as an integration test.

4.5 Phase 2 Timeline

All quantitative results in this section are marked †Phase 2 and will be populated when the following prerequisites are met:

1. NLCD 2021 GeoTIFF downloaded to `data/nlcd/` in the bisect data directory.

2. Zonal statistics computation completed for NC, WI, and TX census tracts using the bisect NLCD pipeline (`bisect fetch -type nlcd`).

3. Land use Parquet files verified against L0 test invariants (Section 3).

4. Single-seed redistricting runs completed under both `land-use` and `geographic` weight modes.

5. LUB and Polsby-Popper metrics computed from plan output.

5 Legal Usage

5.1 Physical Geography and Natural Boundaries in Redistricting Law

The doctrine that redistricting should follow natural geographic boundaries is among the oldest redistricting criteria in American law. Before the one-person-one-vote revolution of the 1960s, courts and legislatures regularly cited rivers, mountain ridges, and coastlines as the appropriate boundaries for legislative districts. After *Reynolds v. Sims* [1964] established the equal population requirement, natural boundaries could no longer govern district shape, but they remained a legitimate *secondary* criterion that plans could follow when doing so was consistent with population equality.

Communities of interest and physical geography. In the modern redistricting framework, physical geography arguments are most commonly presented as a community-of-interest rationale. Residents sharing a physical landscape share environmental concerns (flood risk, wildfire exposure, agricultural water rights), infrastructure needs (coastal access, mountain pass maintenance), and economic dependencies (tourism, agriculture, fishing) that constitute a genuine community of interest for legislative purposes.

The NLCD-based M.2 signal operationalizes this doctrine by treating the physical character of the land cover as a community signal: tracts that share the same development type (residential, commercial, agricultural, forested) share a physical community. Tracts at a development type transition are not a community; they are a boundary.

5.2 Water Boundary Doctrine and Case Law

The hard-cut rule in M.2 (equation 8) implements a well-established legal doctrine: water bodies are natural community boundaries that should not be crossed by district lines without geographic justification.

Judicial recognition of water boundaries. Several redistricting cases have addressed water boundaries directly. In *Carstens v. Lamm*, 543 F. Supp. 68 (D. Colo. 1982), the district court approved a congressional plan that followed the Colorado River as a district boundary, noting that the river constituted a natural geographic division between communities with distinct economic and environmental interests. The court observed that plan proponents could rely on natural geographic features as a neutral, non-partisan rationale for boundary placement.

In *In re Reapportionment of the Colorado General Assembly*, 828 P.2d 185 (Colo. 1992), the Colorado Supreme Court affirmed that natural geographic features, including rivers and ridgelines, are among the legitimate criteria for district boundary placement, and that a plan that follows such features faces a lower burden of justification than one that crosses them arbitrarily.

Outer Banks and barrier island communities. North Carolina's Outer Banks provides a recurring example in redistricting litigation and commission proceedings. The Outer Banks communities—Dare County, Hyde County waterfront, Currituck Banks—are separated from the mainland by Pamlico Sound and Albemarle Sound, bodies of water with no fixed bridge crossings for most of their length. Commission testimony in the 2001, 2011, and 2021 North Carolina redistricting cycles consistently identified Outer Banks communities as a distinct community of interest defined by their physical isolation, coastal economy, and hurricane vulnerability.

Under the M.2 hard-cut rule, Pamlico Sound and Albemarle Sound adjacencies receive zero edge weight, preventing the BISECT algorithm from assigning mainland tracts and Outer Banks tracts to the same district. This directly implements the community-of-interest rationale that North Carolina commissions have accepted in multiple cycles.

Louisiana coastal communities. Louisiana's coastal parishes demonstrate the broader application of the water boundary principle. The bayou network south of New Orleans creates communities (Isle de Jean Charles, Cocodrie, Grand Isle) that are physically accessible only by water or narrow causeways. Redistricting plans that bundle these communities with inland parishes far to the north cross physical barriers that define the coastal community's distinct identity. The NLCD water classification captures the bayou network's dominant land cover character: tracts with majority water classification ($p_{\mathrm{wat}} > 0.5$) along the coastal bayou zone would receive the hard-cut override in any M.2 application to Louisiana.

5.3 The Natural Boundary Doctrine

Beyond water, the broader natural boundary doctrine encompasses any well-defined physical transition that creates meaningfully distinct communities on either side.

Mountain ridgelines. Colorado's redistricting commission has cited the Continental Divide as a natural district boundary in every redistricting cycle since 2001. The Divide separates communities with distinct water rights (Pacific versus Atlantic watershed), ski industry versus ranching economies, and different transportation networks. NLCD captures the mountainous terrain through the forest and shrub/scrub classifications (codes 41–52) that dominate high-altitude tracts. A tract that is 90% evergreen forest ($p_{\mathrm{nat}} \approx 0.90$) has very low cosine similarity with an adjacent Front Range urban tract ($p_{\mathrm{res}} \approx 0.70$), making the transition a natural M.2 cut seam.

Agricultural-suburban boundary. The transition from cultivated crops (NLCD class 82) and pasture (class 81) to low-intensity residential development (class 22) marks the rural-suburban boundary that has been recognized in countless redistricting proceedings as a meaningful community division. The California Citizens Redistricting Commission, in its 2011 and 2021 proceedings, consistently drew district boundaries along the agricultural-suburban edge of the Central Valley, citing the distinct economic and policy interests of farming communities versus suburban commuter communities. The M.2 signal captures this division through the contrast between $p_{\mathrm{nat}} \approx 0.90$ (agricultural tracts) and $p_{\mathrm{res}} \approx 0.70$ (suburban tracts).

5.4 Partisan Neutrality

NLCD data contains no electoral, racial, or demographic information. The land cover classification is derived from satellite imagery and calibrated against ground-truth reference points; it has no knowledge of the voting history or racial composition of the tracts it classifies.

Any partisan effect of M.2 weights—such as the preference for cutting at commercial-residential transitions—is a consequence of the geographic correlation between land use types and partisan voting patterns. This correlation is an empirical fact about the American population landscape (urban commercial cores tend to vote differently from residential suburbs), not a product of the algorithm's choices. Under *Rucho v. Common Cause* [2019], the use of a non-partisan criterion that happens to have partisan effects does not constitute partisan gerrymandering if the criterion is independently justified. Physical land use is an independently justified criterion with deep roots in redistricting practice.

5.5 Expert Witness Deployment Template

Redistricting practitioners deploying M.2 land use weights should structure expert testimony around the following elements:

1. Data provenance declaration. "I used the USGS National Land Cover Database (NLCD) 2021, produced by the Multi-Resolution Land Characteristics Consortium under contract with the U.S. Geological Survey. The NLCD classifies the land surface of the contiguous United States at 30-meter resolution from Landsat satellite imagery. It contains no voter registration data, election results, racial composition data, or any other socioeconomic information about the populations living within the classified areas."

2. Hard-cut rule justification. "The BISECT algorithm assigns a zero-weight edge to any pair of adjacent census tracts where both tracts have majority water classification (more than 50% of their land surface classified as open water by NLCD). This prevents the algorithm from placing such tracts in the same district. This encoding reflects the longstanding redistricting principle that water bodies are natural community boundaries. The water classification is objective and derived from satellite imagery, not from any political or demographic data."

3. Community boundary identification. "In [state], the following physical land use transitions are preserved as district boundaries by the proposed plan: [specific examples, e.g., 'the Pamlico Sound water boundary separates the Outer Banks tracts in Dare County from the mainland, consistent with the commission's recognition of the Outer Banks as a distinct coastal community in its 2011 and 2021 redistricting cycles']."

4. Criterion alignment. "The [state]'s redistricting criteria require preservation of communities defined by geographic and natural boundary considerations. The land use character weights directly implement this criterion: district cuts run preferentially at land use transitions that correspond to recognized natural boundaries between communities."

5.6 When Not to Use M.2

Land use weights should not be used in the following contexts:

- **Dense urban cores.** In highly urbanized census tracts where most of the land is class 24 (high-intensity developed), the M.2 vector becomes near-degenerate ($p_{\text{com}} \approx 1.0$ for all tracts). All tracts look similar under M.2, and the signal adds no information. M.1 (economic character) is more informative in this context.

- **States with no significant natural boundaries.** In states with uniform terrain and no significant water bodies (e.g., Kansas, Iowa), the M.2 signal will produce modest or zero effect. The agricultural uniformity of the Great Plains means most adjacent tracts have similar NLCD vectors; there are no sharp land use transitions to exploit.

- **When hard cuts conflict with VRA compliance.** In rare cases, the hard-cut rule may prevent contiguous assignment of tracts needed for a majority-minority district. In such cases, the hard-cut rule must be overridden via bridge adjacency encoding, and the VRA requirement takes priority.

6 Conclusion

M.2 operationalizes physical land use as a community character signal through the USGS NLCD 2021 land cover classification. The four-component land use vector—residential, commercial, open developed, and natural—captures the physical landscape character of each census tract. Cosine similarity between adjacent tracts modulates edge weights so that METIS cuts preferentially at physical land use transitions. The hard-cut rule for majority-water adjacencies encodes the well-established legal doctrine that water bodies are natural community boundaries.

Together, these design choices address a gap in algorithmic redistricting: the widespread practice of weighting edges by boundary length, with no regard for the physical character of the communities on either side. A long shared boundary between a residential neighborhood and an adjacent commercial corridor is a natural cut seam, not a community bond. M.2 makes this distinction explicit and quantitative.

Relationship to the M-track series. M.2 addresses the physical layer of community character: what is actually built on the ground. It complements M.1 (economic character: what jobs are located there), M.3 (housing character: what kind of housing stock and households occupy it), and M.6 (administrative zone membership: what service districts govern it). These four signals together cover the primary community dimensions that residents identify when they describe why their neighborhood is a recognizable community.

The M.8 composite index [Della-Libera, 2026b] assigns M.2 a default weight of 0.20 (tied with M.1 as the second-highest weight after M.6). This reflects two judgments: land use is a strong community signal (weight 0.20) but slightly weaker than zone co-membership in legal proceedings (weight 0.30), and at least as strong as economic character signals that are better supported by employment data (M.1 at 0.20).

The hard-cut rule as a model for legal defensibility. The water hard-cut rule in M.2 is a model for how legal doctrine can be encoded as an algorithmic constraint. Rather than treating water boundaries as a soft preference, the rule encodes the legal reality: cross-water districts are presumptively invalid and require specific geographic justification (a bridge, a tunnel, an established ferry service with contiguous community use). The zero-weight encoding means the algorithmic output either satisfies the water boundary constraint or is infeasible; there is no gradation between compliance and violation.

Future M-track papers may adopt similar hard-cut rules for other legally established boundaries: county line preservation requirements (where applicable), school district preservation criteria (M.6), and VRA-required minimum threshold adjacencies. The pattern is: when a criterion is binary in legal practice (you may not split this boundary), encode it as a binary constraint in the algorithm, not as a soft weight.

Phase 2 commitments. Phase 2 will complete the following quantitative results, all currently marked †:

1. Land-use boundary coincidence (LUB) metric for NC-14, WI-8, and TX-38, comparing land-use weights to geographic baseline.

2. Hard-cut validation: zero cross-water districts in NC (Pamlico Sound) and TX (Gulf Coast bays) under land-use weights.

3. Polsby-Popper compactness comparison between land-use and geographic weight modes.

4. Multi-seed variance analysis for all three states (Phase 3 if single-seed Phase 2 results are confirmatory).

Phase 2 depends on NLCD 2021 GeoTIFF download and zonal statistics computation. Once the NLCD data is available in the bisect data cache, the full pipeline run can be completed in a single `bisect build` invocation with `-weights-override land-use`.

References

James R Anderson, Ernest E Hardy, John T Roach, and Richard E Witmer. A land use and land cover classification system for use with remote sensor data. Technical Report Professional Paper 964, U.S. Geological Survey, 1976.

Giovanni Della-Libera. M.0: Community character weighting: A framework for operationalizing communities of interest in algorithmic redistricting. Research track M-community-character, 2025.

Giovanni Della-Libera. M.3: Housing character edge weights via ACS. Research track M-community-character, 2026a.

Giovanni Della-Libera. M.8: Composite community character index: Synthesizing economic, physical, administrative, and transit signals. Research track M-community-character, 2026b.

Jon Dewitz and U.S. Geological Survey. National land cover database (NLCD) 2021 products. U.S. Geological Survey data release, 2023.

Collin Homer, Jon Dewitz, Suming Jin, George Xian, Catherine Costello, Patrick Danielson, Leila Gass, Michelle Funk, James Wickham, Stephen Stehman, Roger Auch, and Kurt Riitters. Conterminous United States land cover change patterns 2001–2016 from the 2016 National Land Cover Database. *ISPRS Journal of Photogrammetry and Remote Sensing*, 162:184–199, 2020. doi: 10.1016/j.isprsjprs.2020.02.019. MRLC validation methodology reference; classification accuracy protocols apply to 2021 NLCD release under same framework.

Daniel D Polsby and Robert D Popper. The third criterion: Compactness as a procedural safeguard against partisan gerrymandering. *Yale Law & Policy Review*, 9(2):301–353, 1991.

Joseph E Schwartzberg. Reapportionment, gerrymanders, and the notion of "compactness". *Minnesota Law Review*, 50:443, 1966.

U.S. Census Bureau. TIGER/Line shapefiles, 2020. `https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html`, 2020.

U.S. Supreme Court. *Reynolds v. Sims*, 377 u.s. 533 (1964), 1964.

U.S. Supreme Court. *Rucho v. Common Cause*, 588 u.s. 684 (2019), 2019.